FORMULA 1 RACE INTELLIGENCE DASHBOARD

Project Report

Project Developer: Gyanvi Jain
Programme: B.Tech – First Year
Project Type: Individual Project
Technology: Tableau, Excel, Python, Flask, HTML, CSS

1. INTRODUCTION

1.1 Project Overview

The Formula 1 Race Intelligence Dashboard is a data visualisation and analysis project designed to explore Formula 1 racing data across multiple seasons. It presents key performance indicators, driver rankings, team performance, race outcomes, circuit locations, and reliability insights through interactive visualisations.

1.2 Purpose

The purpose of this project is to simplify the analysis of Formula 1 data by transforming raw racing records into meaningful visual insights. It helps users compare drivers and teams, examine race outcomes, identify DNF (Did Not Finish) patterns, and explore performance trends across seasons.

2. IDEATION PHASE

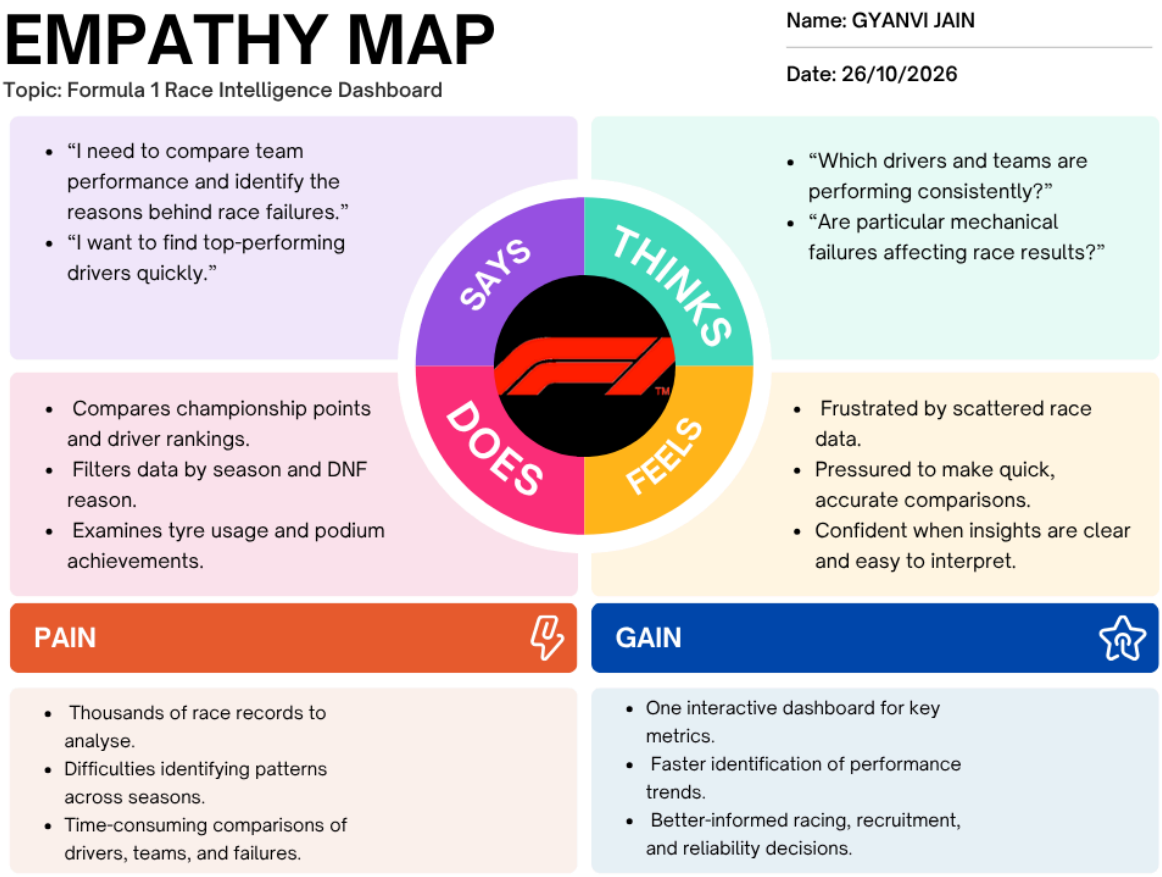
2.1 Problem Statement

Formula 1 generates large amounts of data related to drivers, teams, circuits, race results, and vehicle reliability. Analysing these records manually can be time-consuming and makes comparisons across seasons difficult. A centralised, interactive dashboard is needed to present this information clearly and support data-driven analysis.

2.2 Empathy Map Canvas

The intended users include Formula 1 enthusiasts, sports analysts, and viewers interested in understanding driver and team performance.

- **Says:** “I want to compare drivers and teams across seasons.”
- **Thinks:** “Which drivers and teams perform consistently well?”
- **Does:** Examines rankings, race results, and championship statistics.
- **Feels:** Interested but may find raw datasets difficult to interpret.
- **Pains:** Scattered information, complex data, and time-consuming comparisons.
- **Gains:** Clear visualisations, interactive filters, and quicker insights.



2.3 Brainstorming

The following ideas were considered during ideation:

1. Interactive driver performance leaderboard using cumulative points.
2. DNF analysis by driver, team, and reason.

3. Tyre compound usage and strategy analysis.
4. Team championship comparisons across seasons.
5. Geographic analysis of circuits and driver participation.
6. Podium achievements and seasonal performance trends.

These ideas informed the selection of interactive charts, KPI cards, filters, and analytical views.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The user journey consists of the following stages:

1. **Discover:** Opens the project website or Tableau dashboard.
2. **Explore:** Reviews KPI cards and overall Formula 1 statistics.
3. **Filter:** Selects a season, DNF reason, or Top N value.
4. **Analyse:** Compares driver rankings, team performance, and reliability.
5. **Interpret:** Identifies trends and draws conclusions from the visualisations.

3.2 Solution Requirement

Functional requirements

- Display key performance indicators.
- Show driver rankings and points-based comparisons.
- Analyse DNF reasons and driver reliability.
- Compare team performance across seasons.
- Present circuit locations and geographic insights.
- Provide interactive filters and a Top N parameter.

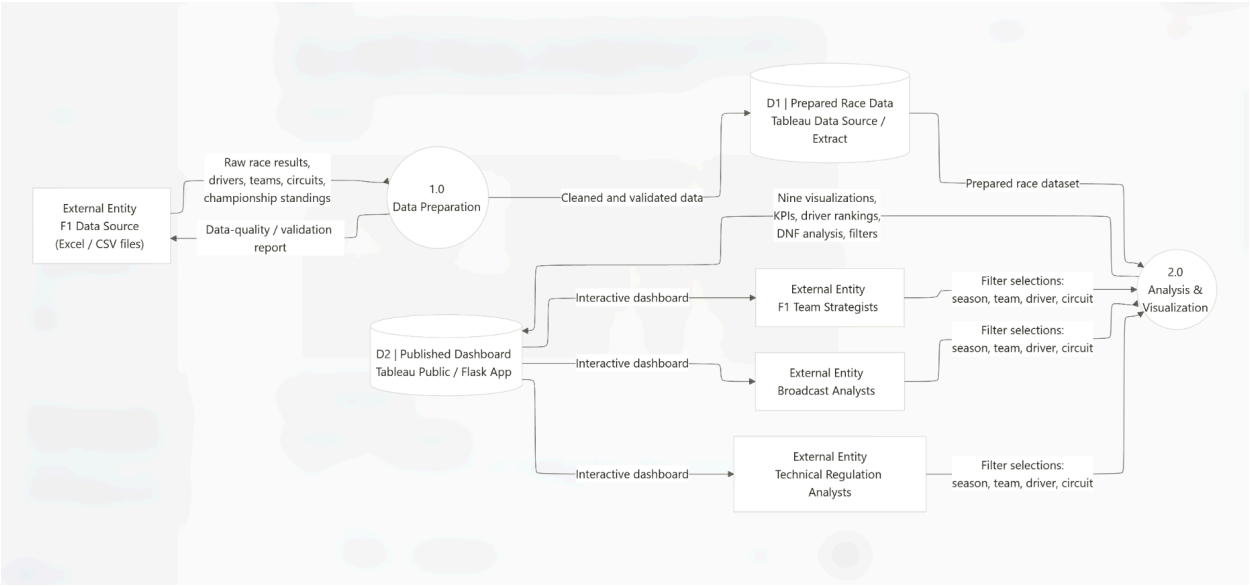
Non-functional requirements

- Clear and readable visualisations.
- Reliable calculations and consistent data.
- Responsive website layout.
- Reasonable loading performance.
- Easy navigation and usability.

3.3 Data Flow Diagram

The project follows this basic data flow:

Formula 1 Dataset → Data Preparation → Tableau Data Model → Calculations and Filters → Visualisations → User Insights



The dataset is prepared in Excel and connected to Tableau for analysis. The resulting dashboards and story are published on Tableau Public and can be displayed through the project website.

3.4 Technology Stack

Technology	Purpose
Microsoft Excel	Dataset storage and preparation
Tableau Desktop	Data analysis and dashboard development
Tableau Public	Publishing dashboards and stories
Python	Website backend
Flask	Web application framework
HTML	Website structure
CSS	Website styling and responsive layout
GitHub	Source code and project documentation hosting

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project addresses the difficulty of understanding large Formula 1 datasets by providing a centralised and interactive visualisation system. KPI cards summarise important statistics, while charts and filters allow users to investigate drivers, teams, circuits, and race outcomes.

4.2 Proposed Solution

The proposed solution is an interactive Formula 1 Race Intelligence Dashboard supported by a Flask-based project website. The dashboard includes:

- KPI cards for teams, seasons, race outcomes, and circuits.
- Driver performance leaderboard.
- DNF and reliability analysis.
- Team championship comparisons.
- Tyre compound and race analysis.
- Geographic and podium-related visualisations.

4.3 Solution Architecture

The system consists of four main layers:

1. **Data Layer:** Formula 1 data stored in Excel.
2. **Analysis Layer:** Tableau data preparation, calculations, filters, and visualisations.
3. **Presentation Layer:** Published Tableau dashboards and story.
4. **Web Layer:** Flask, HTML, and CSS website that presents project information and embeds Tableau views.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned in stages to organise development and testing.

Phase	Activities
Ideation	Identify the problem and brainstorm possible solutions.

Requirement Analysis	Define users, functional requirements, and technology needs.
Data Preparation	Connect the dataset and check data types and missing values.
Dashboard Development	Build KPI cards, charts, filters, and calculated fields.
Website Development	Create the Flask website and integrate published Tableau views.
Testing	Check calculations, filters, visualisations, and website functionality.
Deployment & Documentation	Publish the dashboard, maintain source code on GitHub, and prepare the report.

The project backlog and sprint schedule can be included as supporting evidence in this section.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The following tests should be performed and their actual outcomes recorded before submission.

Test Case	Expected Result	Actual Result / Status
Dashboard loading	Dashboard loads correctly.	Recorded
KPI verification	KPI values match the underlying data.	Recorded
Season filter	Visualisations update according to the selected season.	Recorded
DNF reason filter	Relevant reliability results are displayed.	Recorded
Top N parameter	Leaderboard displays the selected number of drivers.	Recorded
Website navigation	Navigation links open the correct sections.	Recorded

Tableau embedding	Published views appear correctly on the website.	Recorded

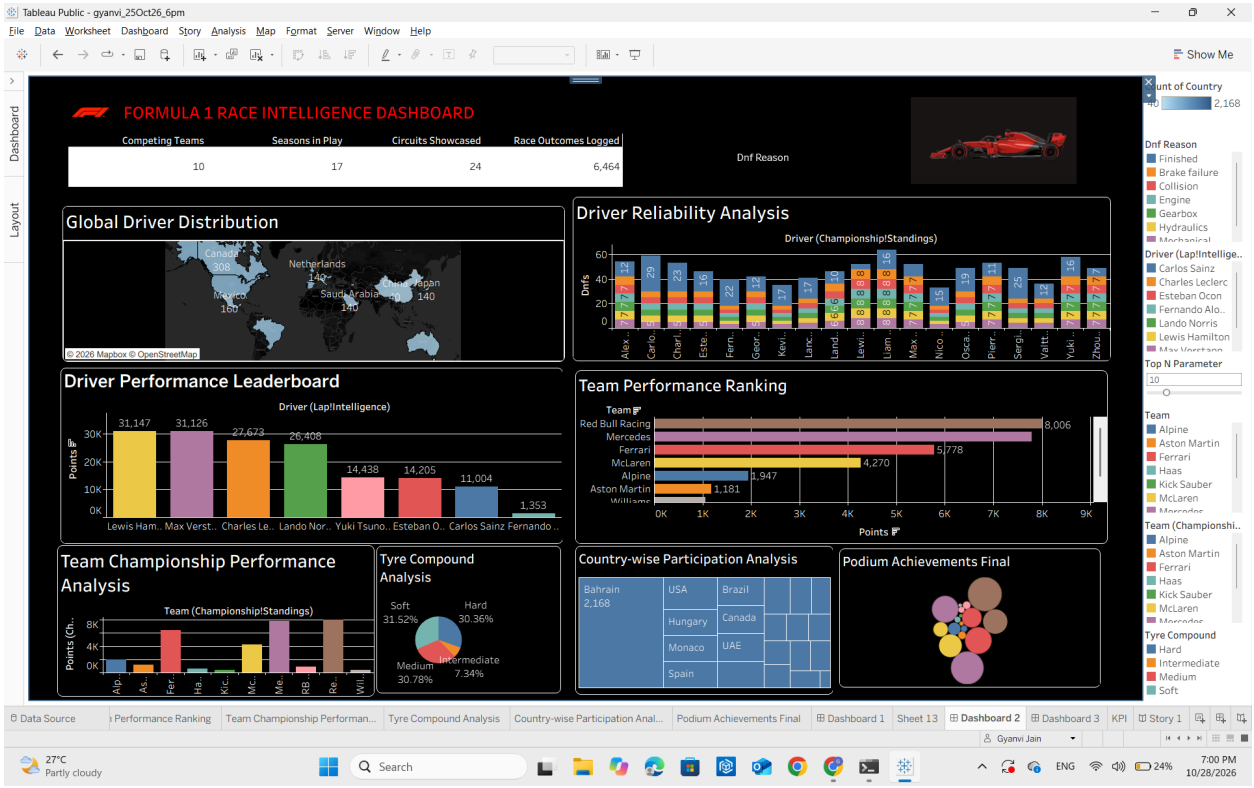
Record measured loading times and observed errors if performance measurements are required. Do not report tests as passed until they have been performed.

7. RESULTS

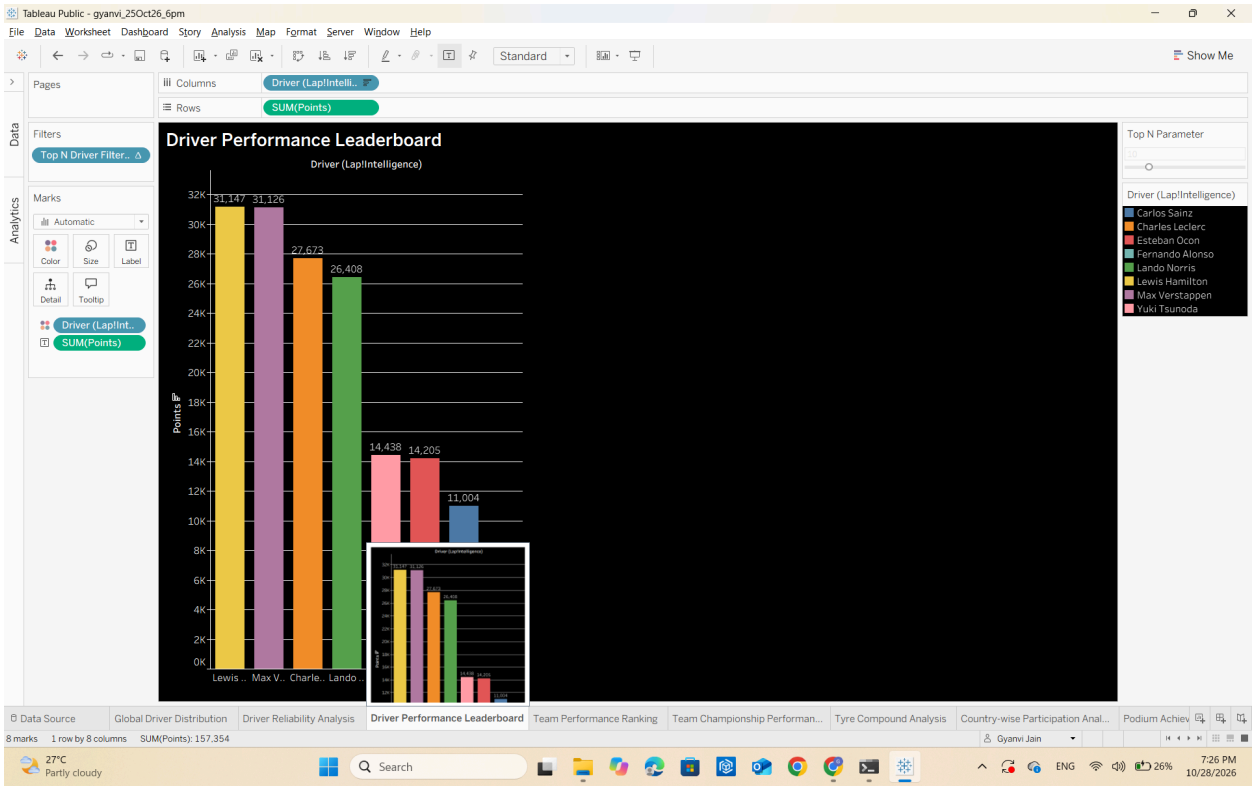
7.1 Output Screenshots

Include screenshots of the completed project, with a short caption for each:

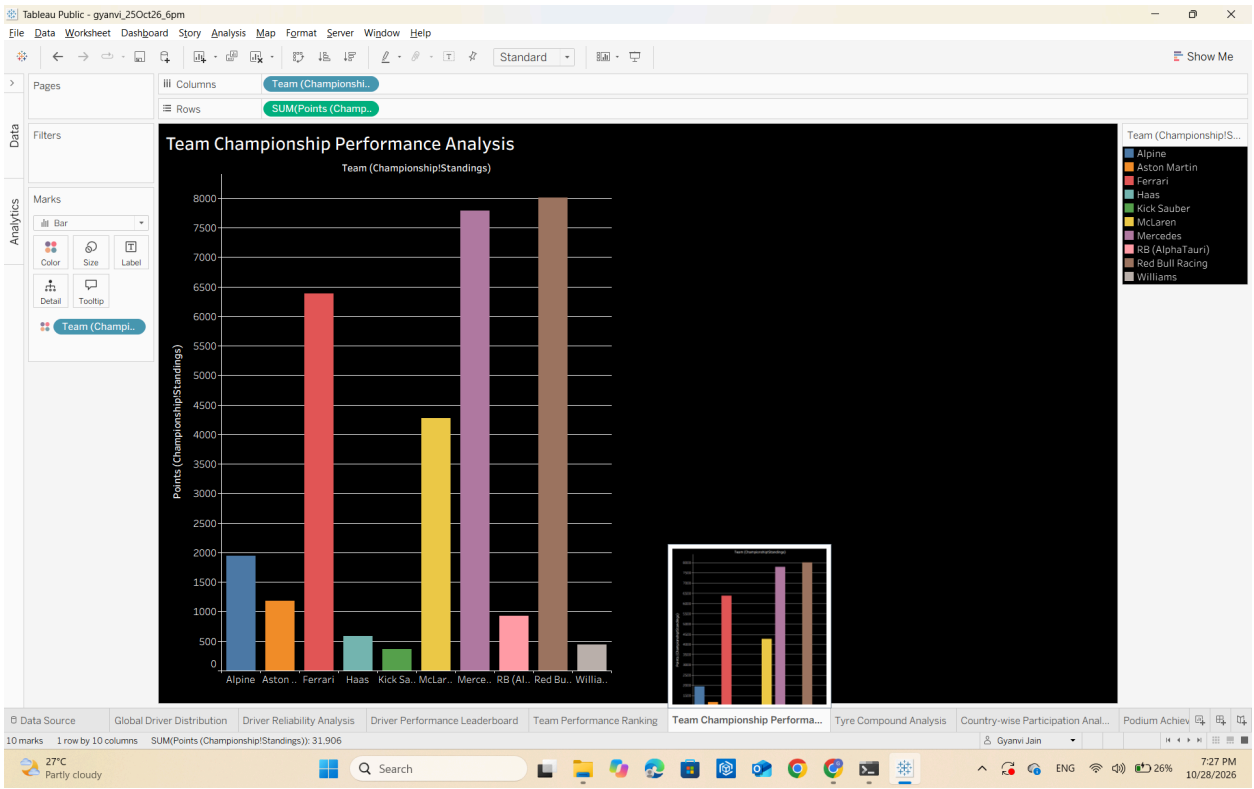
1. Formula 1 Race Intelligence Dashboard overview.
2. **Figure 2:** KPI cards and driver performance leaderboard.



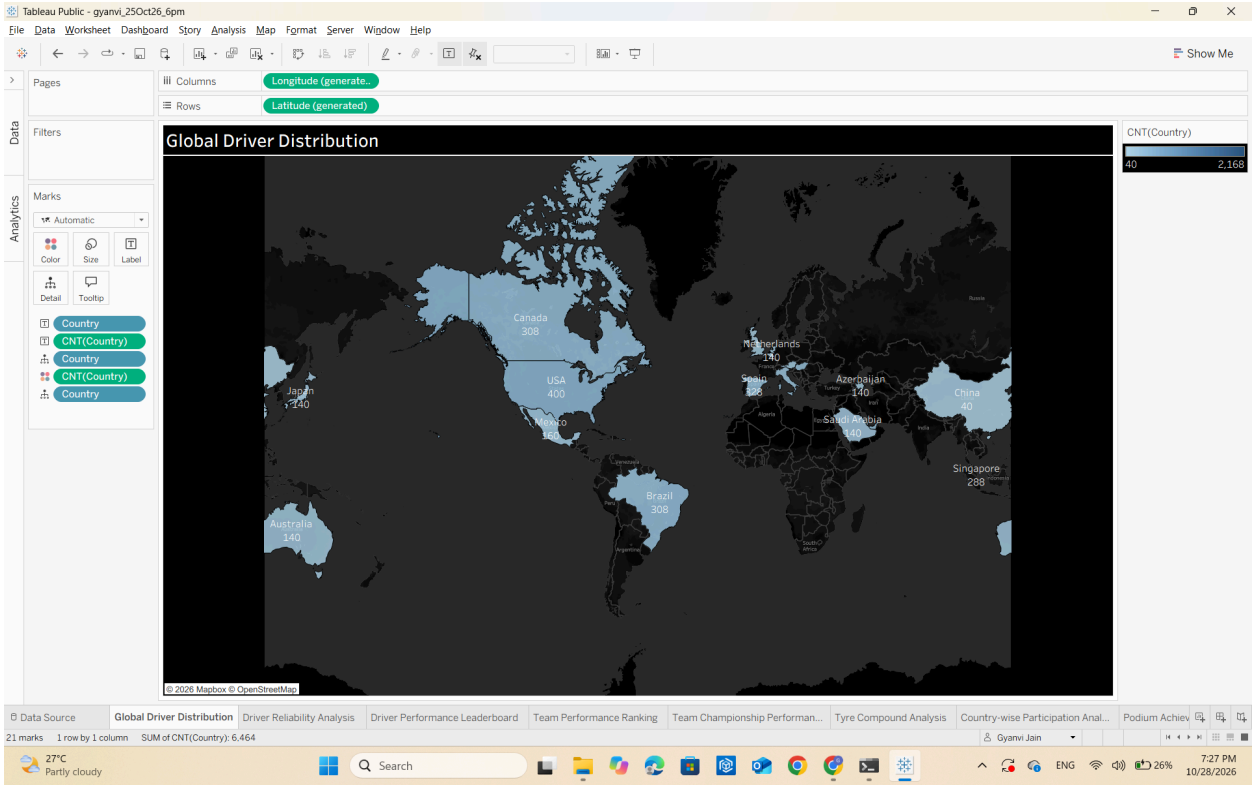
3. **Figure 3:** Driver reliability and DNF analysis.



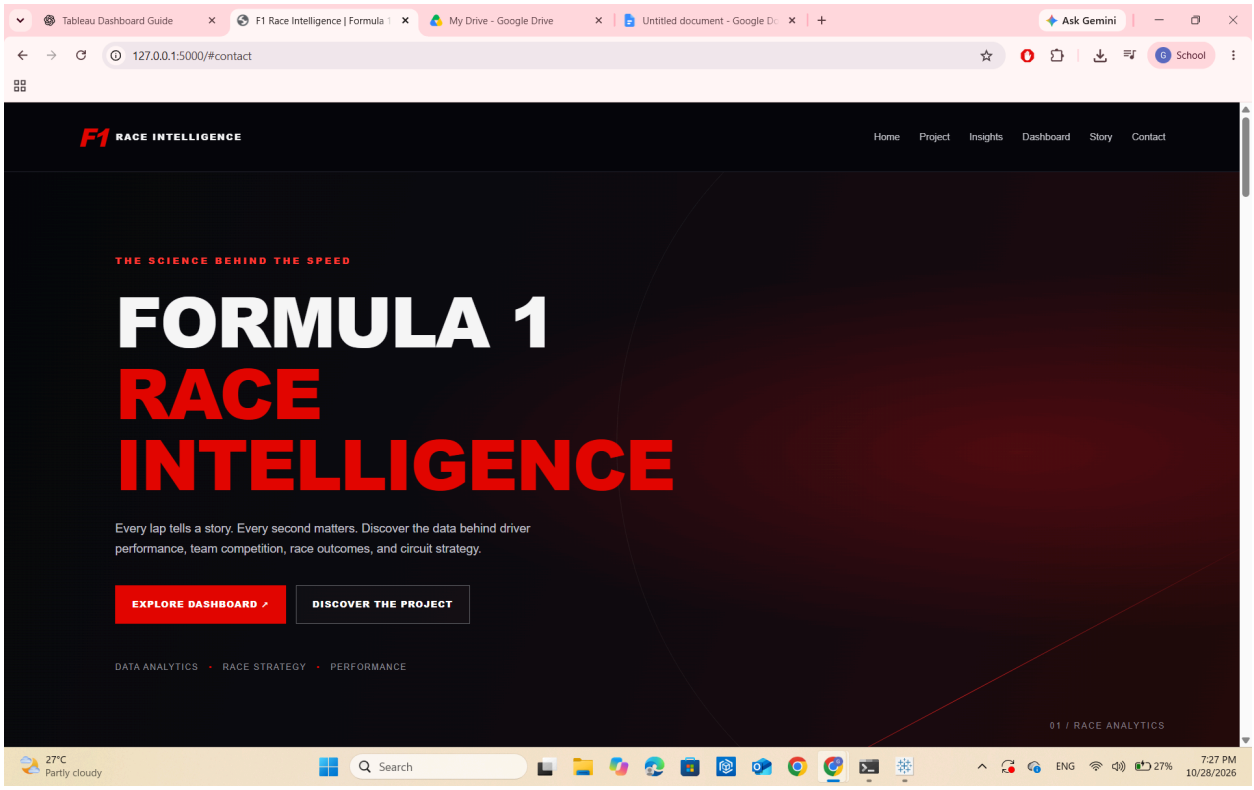
4. **Figure 4:** Team championship comparison.

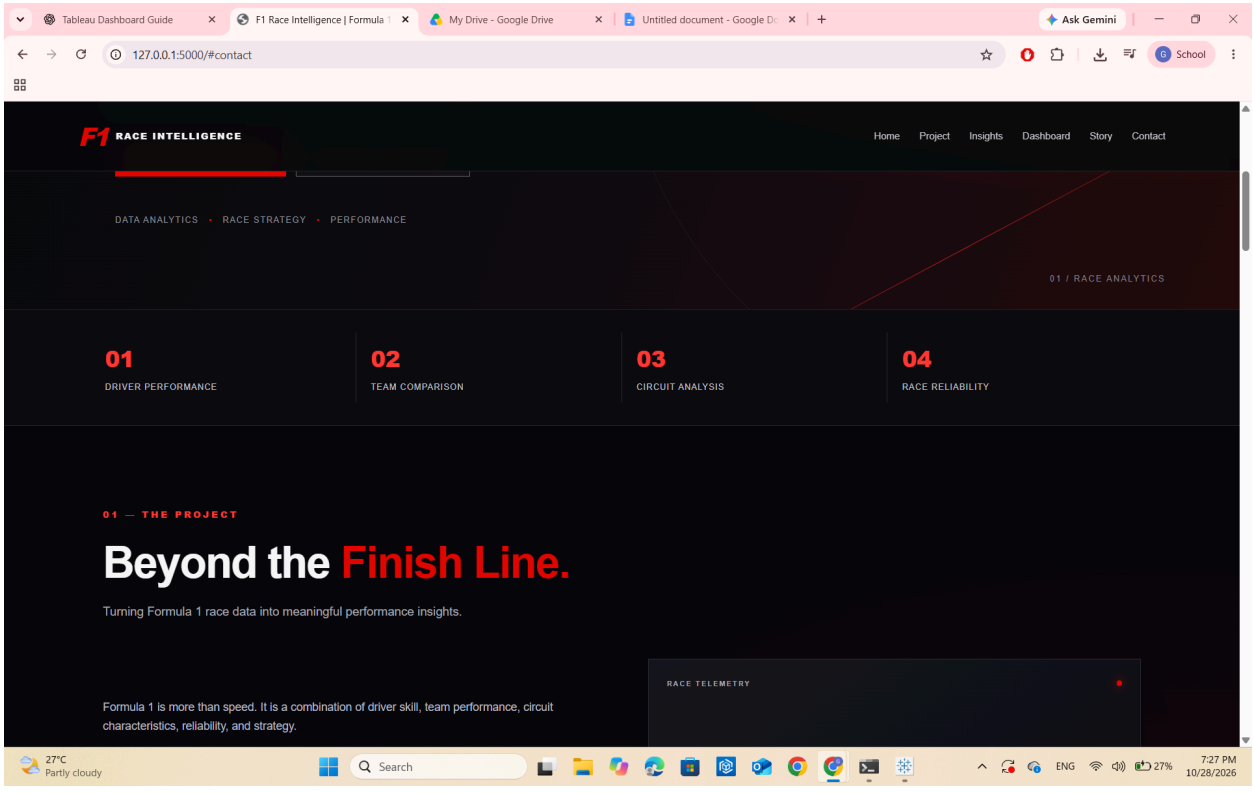


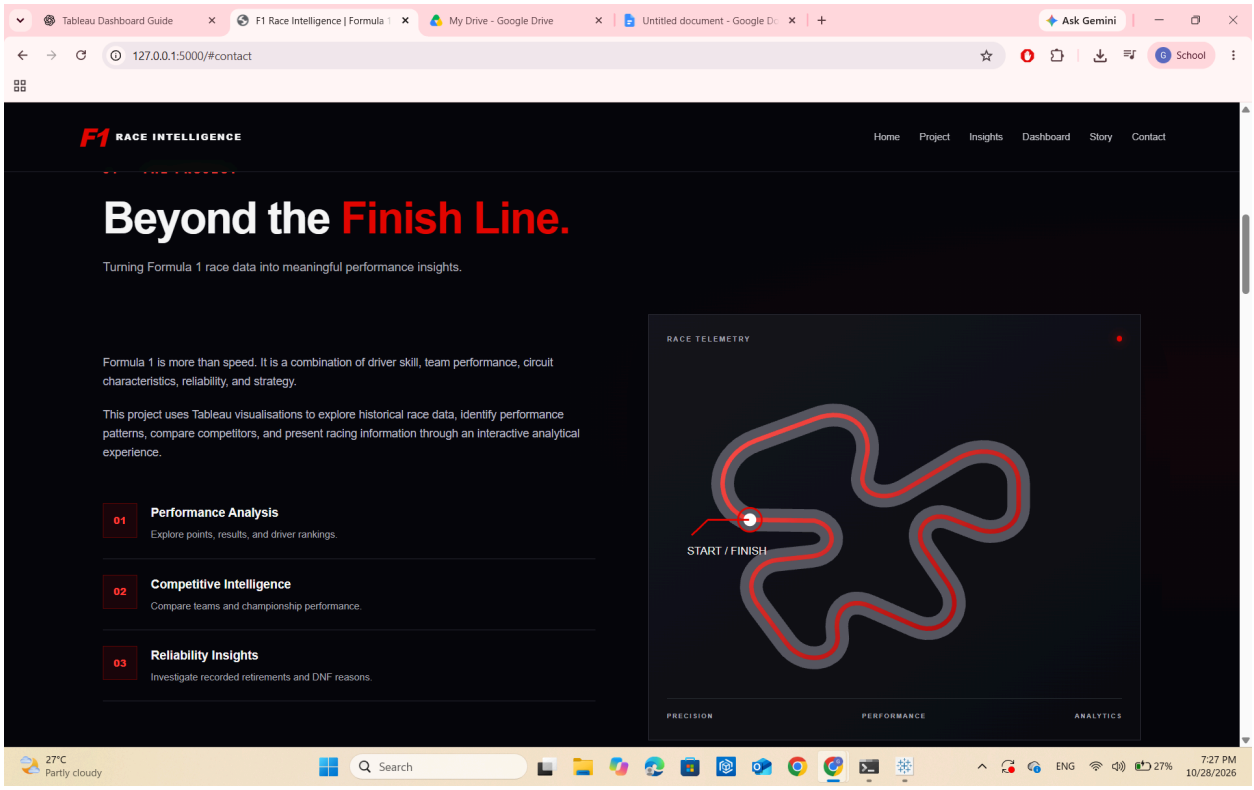
5. **Figure 5:** Geographic participation or circuit map.

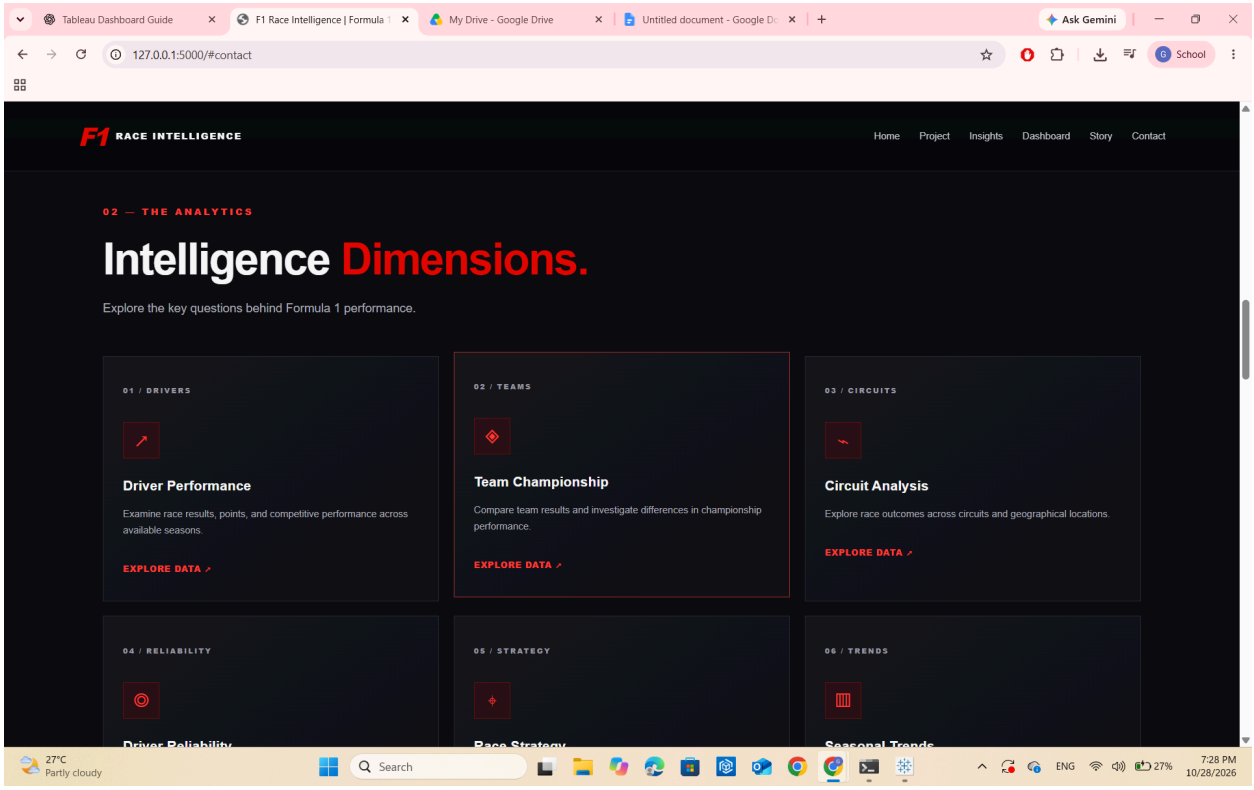


6. **Figure 6:** Formula 1 project website.


Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google D
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
THE SCIENCE BEHIND THE SPEED
FORMULA 1
RACE
INTELLIGENCE
Every lap tells a story. Every second matters. Discover the data behind driver performance, team competition, race outcomes, and circuit strategy.
EXPLORE DASHBOARD ↗
DISCOVER THE PROJECT
DATA ANALYTICS · RACE STRATEGY · PERFORMANCE
01 / RACE ANALYTICS
27°C
Partly cloudy
Search
ENG
27%
7:27 PM
10/28/2026


Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google D
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
DATA ANALYTICS · RACE STRATEGY · PERFORMANCE
01 / RACE ANALYTICS
01
DRIVER PERFORMANCE
02
TEAM COMPARISON
03
CIRCUIT ANALYSIS
04
RACE RELIABILITY
01 — THE PROJECT
Beyond the Finish Line.
Turning Formula 1 race data into meaningful performance insights.
RACE TELEMETRY
Formula 1 is more than speed. It is a combination of driver skill, team performance, circuit characteristics, reliability, and strategy.
27°C
Partly cloudy
Search
ENG
27%
7:27 PM
10/28/2026


Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google D
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
Beyond the Finish Line.
Turning Formula 1 race data into meaningful performance insights.
Formula 1 is more than speed. It is a combination of driver skill, team performance, circuit characteristics, reliability, and strategy.
This project uses Tableau visualisations to explore historical race data, identify performance patterns, compare competitors, and present racing information through an interactive analytical experience.
01
Performance Analysis
Explore points, results, and driver rankings.
02
Competitive Intelligence
Compare teams and championship performance.
03
Reliability Insights
Investigate recorded retirements and DNF reasons.
RACE TELEMETRY
START / FINISH
PRECISION
PERFORMANCE
ANALYTICS
27°C
Partly cloudy
Search
ENG
27%
7:27 PM
10/28/2026


Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google D
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
02 — THE ANALYTICS
Intelligence Dimensions.
Explore the key questions behind Formula 1 performance.
01 / DRIVERS
Driver Performance
Examine race results, points, and competitive performance across available seasons.
EXPLORE DATA ↗
02 / TEAMS
Team Championship
Compare team results and investigate differences in championship performance.
EXPLORE DATA ↗
03 / CIRCUITS
Circuit Analysis
Explore race outcomes across circuits and geographical locations.
EXPLORE DATA ↗
04 / RELIABILITY
05 / STRATEGY
06 / TRENDS
27°C
Partly cloudy
Search
ENG
27%
7:28 PM
10/28/2026

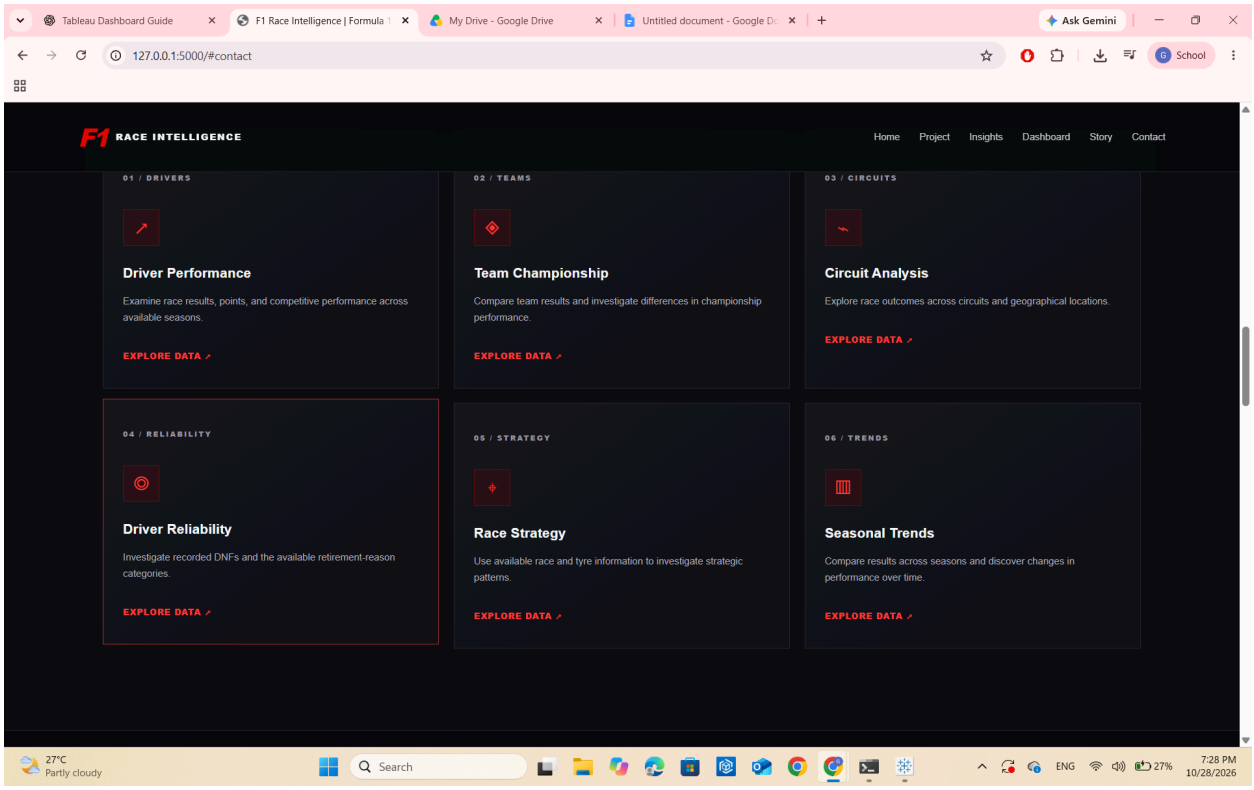
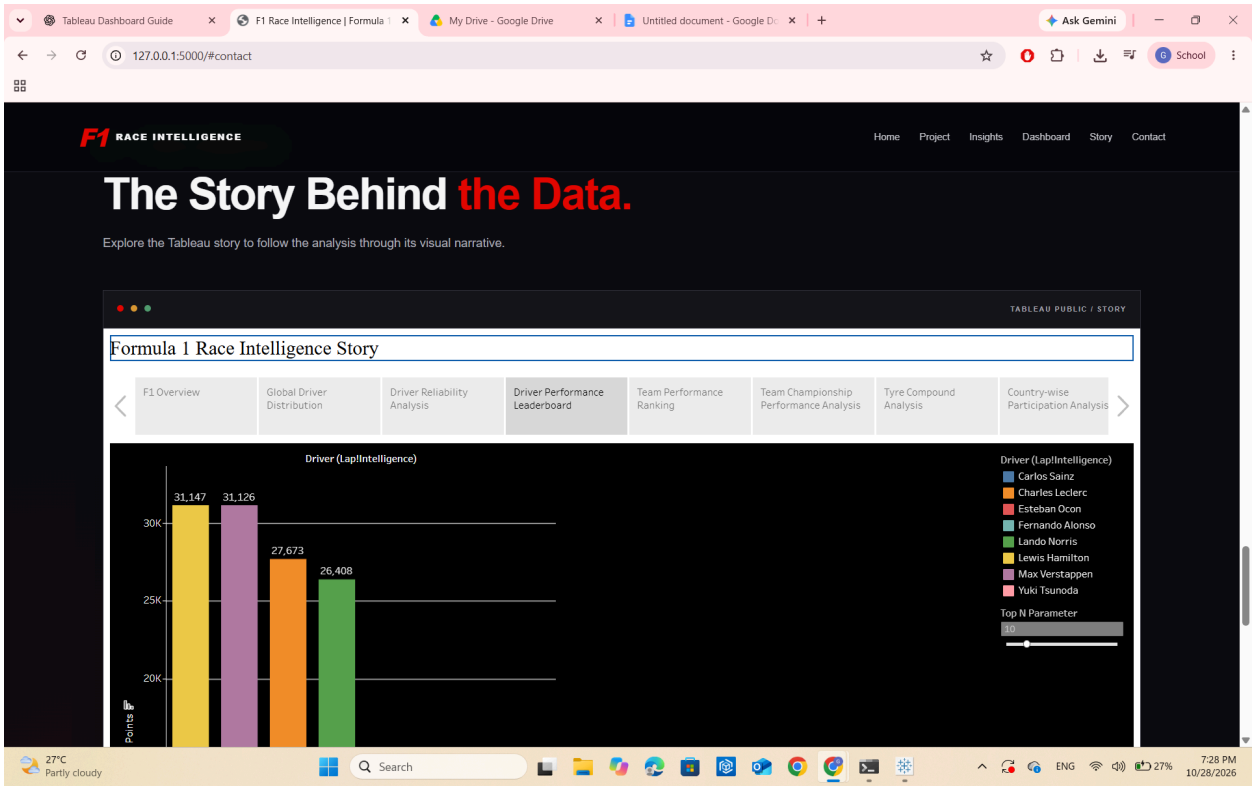
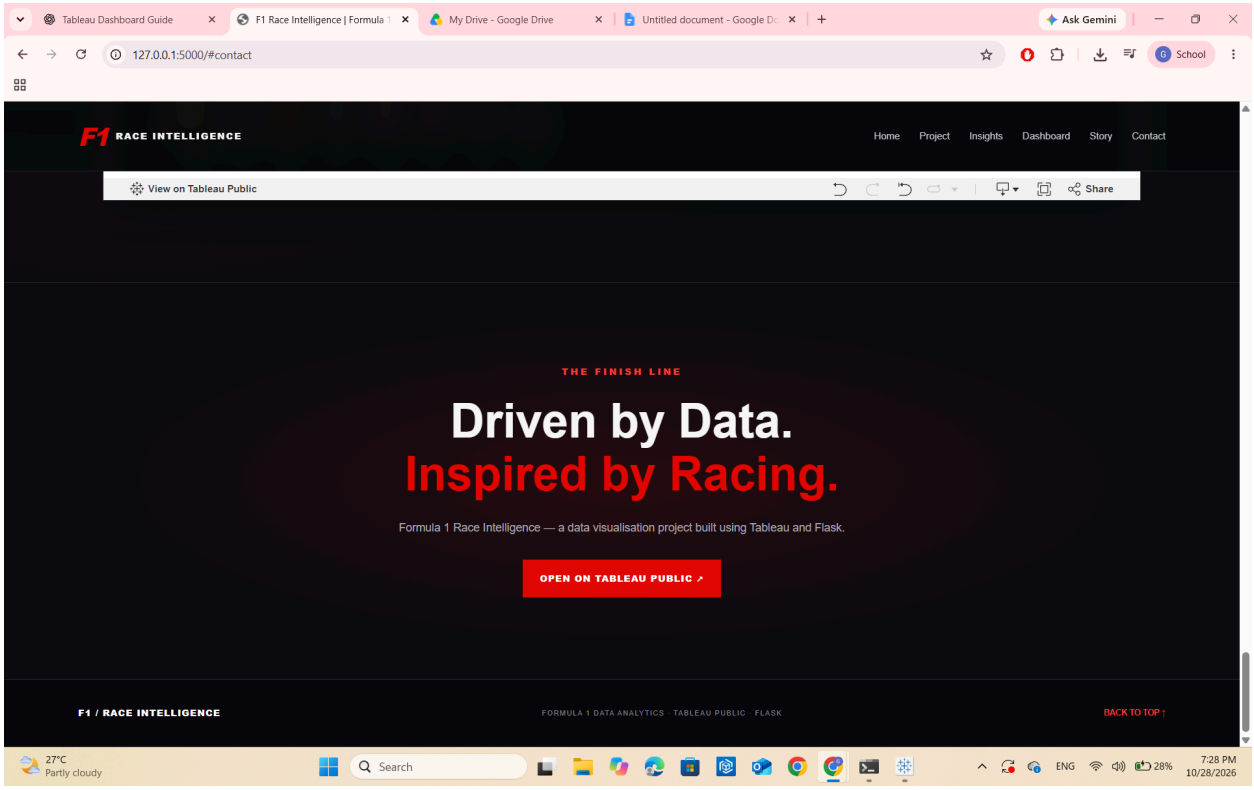

Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google Do
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
01 / DRIVERS
Driver Performance
Examine race results, points, and competitive performance across available seasons.
EXPLORE DATA ↗
02 / TEAMS
Team Championship
Compare team results and investigate differences in championship performance.
EXPLORE DATA ↗
03 / CIRCUITS
Circuit Analysis
Explore race outcomes across circuits and geographical locations.
EXPLORE DATA ↗
04 / RELIABILITY
Driver Reliability
Investigate recorded DNFs and the available retirement-reason categories.
EXPLORE DATA ↗
05 / STRATEGY
Race Strategy
Use available race and tyre information to investigate strategic patterns.
EXPLORE DATA ↗
06 / TRENDS
Seasonal Trends
Compare results across seasons and discover changes in performance over time.
EXPLORE DATA ↗
27°C
Partly cloudy
Search
ENG
27%
7:28 PM
10/28/2026


Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google Do
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
Global Driver Distribution
Canada 308
Netherlands 148
Japan 140
Mexico 160
Saudi Arabia 140
China 40
Singapo 288
© 2026 Mapbox © OpenStreetMap
Driver Reliability Analysis
Driver (Championship!Standings)
Drnfs
Driver Performance Leaderboard
Driver (Lap!Intelligence)
Points
31,147
31,126
27,673
26,408
14,438
14,205
11,004
1,353
Team Performance Ranking
Team
Red Bull Racing 8,006
Mercedes 7,790
Ferrari 5,778
McLaren 4,270
Alpine 1,947
Aston Martin 1,181
Points
Team Championship Performance Analysis
Team (Championship!Standings)
Tyre Compound Analysis
Soft 31.52%
Hard 30.36%
Medium 30.78%
Intermedia 7.34%
Country-wise Participation Analysis
Bahrain 2,168
UAE
Podium Achievements Final
Collision
Engine
Gearbox
Hydraulics
Carlos Sainz
Charles Leclerc
Esteban Ocon
Fernando Alonso
Lando Norris
Lewis Hamilton
Top N Parameter
Alpine
Aston Martin
Ferrari
Haas
Kick Sauber
McLaren
Hard
Intermediate
Medium
Soft
27°C
Partly cloudy
Search
ENG
27%
7:28 PM
10/28/2026


Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google D
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
The Story Behind the Data.
Explore the Tableau story to follow the analysis through its visual narrative.
TABLEAU PUBLIC / STORY
Formula 1 Race Intelligence Story
F1 Overview
Global Driver Distribution
Driver Reliability Analysis
Driver Performance Leaderboard
Team Performance Ranking
Team Championship Performance Analysis
Tyre Compound Analysis
Country-wise Participation Analysis
Driver (Lap!ntelligence)
31,147
31,126
27,673
26,408
30K
25K
20K
Points
Driver (Lap!ntelligence)
Carlos Sainz
Charles Leclerc
Esteban Ocon
Fernando Alonso
Lando Norris
Lewis Hamilton
Max Verstappen
Yuki Tsunoda
Top N Parameter
10
27°C
Partly cloudy
Search
ENG
27%
7:28 PM
10/28/2026


Tableau Dashboard Guide
F1 Race Intelligence | Formula
My Drive - Google Drive
Untitled document - Google D
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
View on Tableau Public
Share
THE FINISH LINE
Driven by Data.
Inspired by Racing.
Formula 1 Race Intelligence — a data visualisation project built using Tableau and Flask.
OPEN ON TABLEAU PUBLIC ↗
F1 / RACE INTELLIGENCE
FORMULA 1 DATA ANALYTICS · TABLEAU PUBLIC · FLASK
BACK TO TOP ↑
27°C
Partly cloudy
Search
ENG
28%
7:28 PM
10/28/2026

8. ADVANTAGES & DISADVANTAGES

Advantages

- Converts complex racing data into understandable visualisations.
- Enables comparisons between drivers and teams.
- Provides interactive filtering and ranking.
- Highlights DNF patterns and performance trends.
- Makes the dashboard accessible through Tableau Public.
- Provides a dedicated website for project presentation.

Disadvantages

- Analysis depends on the completeness and accuracy of the dataset.
- Some insights require familiarity with Formula 1 terminology.
- Dashboard performance may depend on network connectivity and dataset size.
- Results are limited to the variables and seasons available in the dataset.
- Tableau Public dashboards generally require an internet connection to access published content.

9. CONCLUSION

The Formula 1 Race Intelligence Dashboard demonstrates how data visualisation can simplify the analysis of motorsport data. By combining KPI cards, interactive charts, filters, and driver and team comparisons, the project provides a structured way to explore Formula 1 performance and race outcomes. The supporting website improves the presentation of the dashboard and provides a central location for project information.

10. FUTURE SCOPE

Future enhancements may include:

- Adding live or regularly updated Formula 1 data.
- Developing more detailed tyre strategy and pit-stop analysis.
- Adding predictive models for race outcomes and driver performance.
- Improving mobile responsiveness and accessibility.
- Adding more detailed season-by-season and circuit-level comparisons.
- Introducing automated data refresh and advanced analytics.

11. APPENDIX

A. Source Code

GitHub Repository:
https://github.com/GyanviJ/Formula-1-Race-Intelligence-Dashboard

B. Dataset Link

Given in skillwallet

C. Tableau Dashboard

https://public.tableau.com/app/profile/gyanvi.jain/viz/gyanvi_25Oct26_6pm/Dashboard2?publish=yes

D. Tableau Story

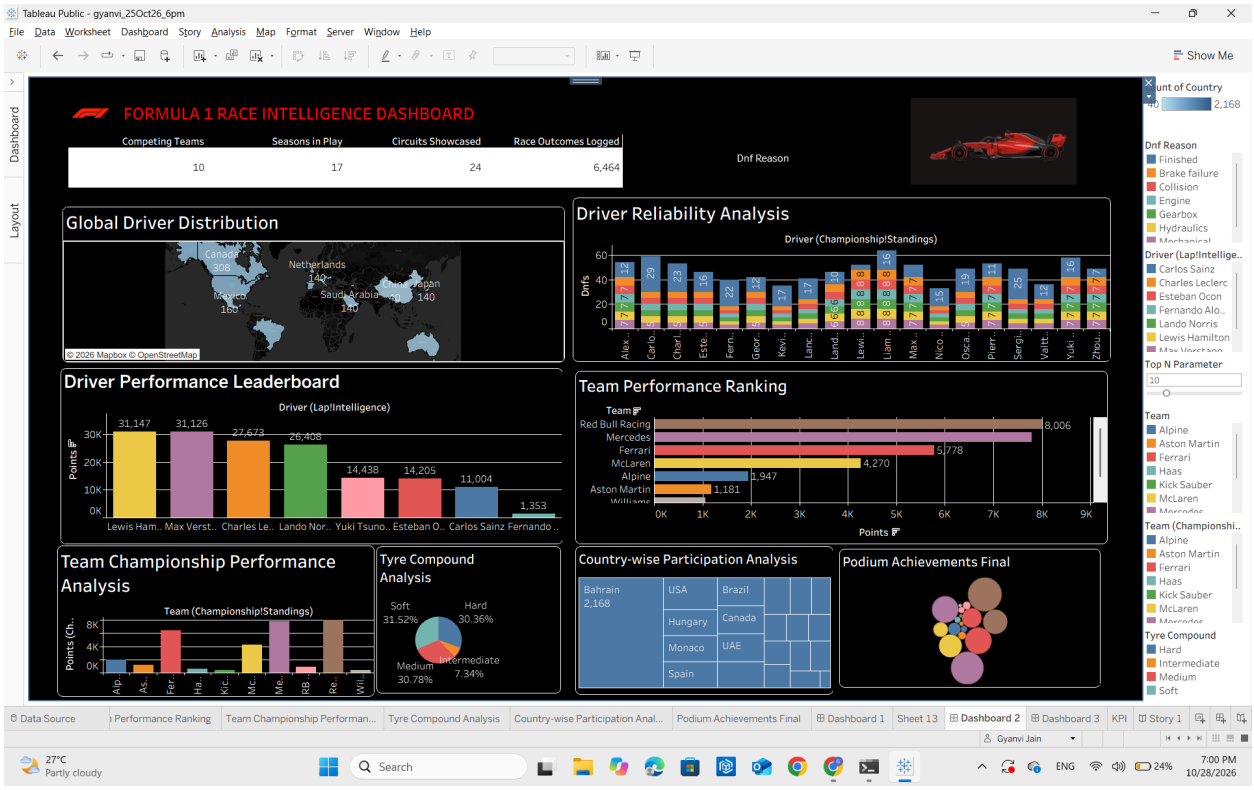
https://public.tableau.com/app/profile/gyanvi.jain/viz/gyanvi_25Oct26_6pm/Story1?publish=yes

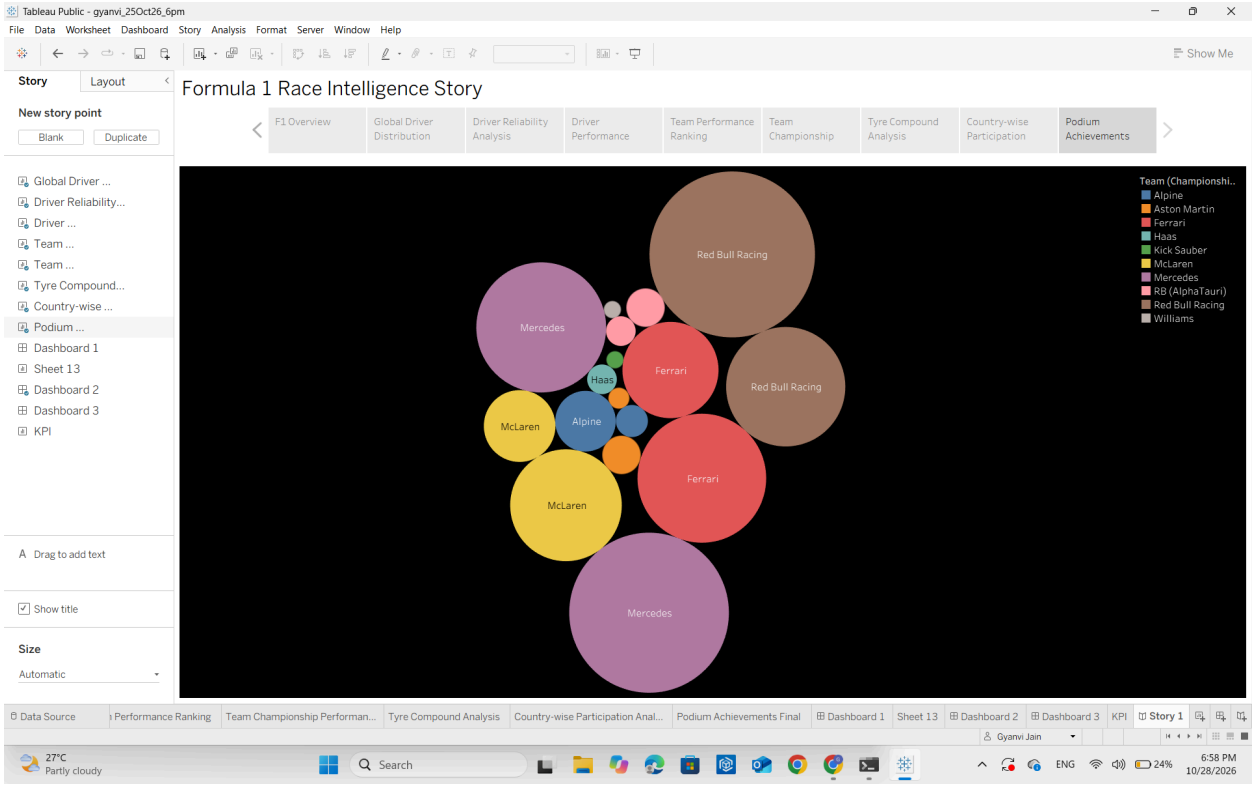
E. Project Demo Link

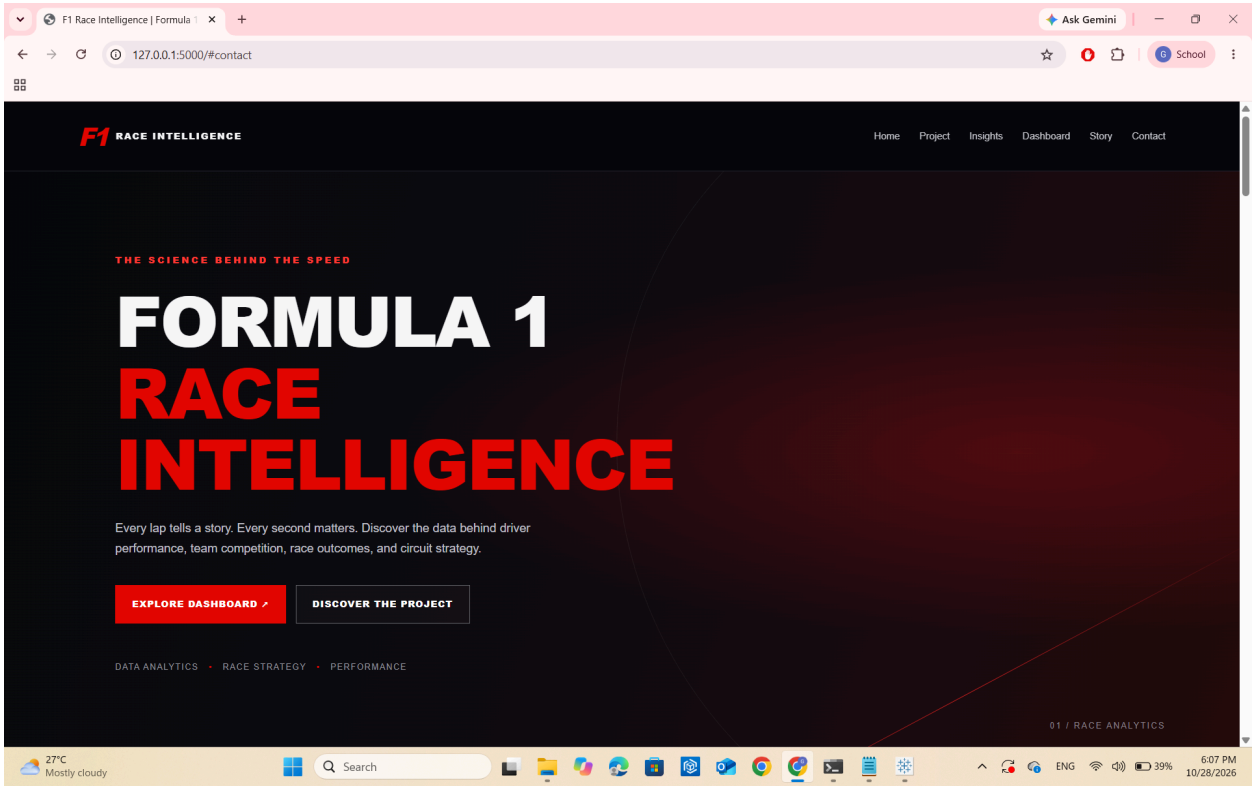
https://drive.google.com/file/d/1bQTKXPDJs-PxYVVCts5FbNybcHYAXa6M/view?usp=sharing

F. Supporting Documents

(Attached below)


Tableau Public - gyanvi_25Oct26_6pm
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Show Me
Dashboard
Layout
FORMULA 1 RACE INTELLIGENCE DASHBOARD
Competing Teams
Seasons in Play
Circuits Showcased
Race Outcomes Logged
10
17
24
6,464
Dnf Reason
Global Driver Distribution
Canada 308
Netherlands 140
Mexico 160
Saudi Arabia 140
China 140
Japan 140
© 2026 Mapbox © OpenStreetMap
Driver Reliability Analysis
Driver (Championship!Standings)
Dnfs
Driver Performance Leaderboard
Driver (Lap!Intelligence)
Points
31,147
31,126
27,673
26,408
14,438
14,205
11,004
1,353
Lewis Ham..
Max Verst..
Charles Le..
Lando Nor..
Yuki Tsuno..
Esteban O..
Carlos Sainz
Fernando ..
Team Performance Ranking
Team
Red Bull Racing 8,006
Mercedes
Ferrari 5,778
McLaren 4,270
Alpine 1,947
Aston Martin 1,181
Points
Team Championship Performance Analysis
Team (Championship!Standings)
Points (Ch..
Tyre Compound Analysis
Soft 31.52%
Hard 30.36%
Intermediate 7.34%
Medium 30.78%
Country-wise Participation Analysis
Bahrain 2,168
USA
Brazil
Hungary
Canada
Monaco
UAE
Spain
Podium Achievements Final
Count of Country
2,168
Dnf Reason
Finished
Brake failure
Collision
Engine
Gearbox
Hydraulics
Driver (Lap!Intellige..
Carlos Sainz
Charles Leclerc
Esteban Ocon
Fernando Alo..
Lando Norris
Lewis Hamilton
Top N Parameter
10
Team
Alpine
Aston Martin
Ferrari
Haas
Kick Sauber
McLaren
Team (Championshi..
Tyre Compound
Hard
Intermediate
Medium
Soft
Data Source
Performance Ranking
Team Championship Performan...
Tyre Compound Analysis
Country-wise Participation Anal...
Podium Achievements Final
Dashboard 1
Sheet 13
Dashboard 2
Dashboard 3
KPI
Story 1
Gyanvi Jain
27°C
Partly cloudy
Search
ENG
24%
7:00 PM
10/28/2026


Tableau Public - gyanvi_25Oct26_6pm
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Show Me
Story
Layout
New story point
Blank
Duplicate
Global Driver ...
Driver Reliability...
Driver ...
Team ...
Team ...
Tyre Compound...
Country-wise ...
Podium ...
Dashboard 1
Sheet 13
Dashboard 2
Dashboard 3
KPI
Drag to add text
Show title
Size
Automatic
Formula 1 Race Intelligence Story
F1 Overview
Global Driver Distribution
Driver Reliability Analysis
Driver Performance
Team Performance Ranking
Team Championship
Tyre Compound Analysis
Country-wise Participation
Podium Achievements
Team (Championshi..
Alpine
Aston Martin
Ferrari
Haas
Kick Sauber
McLaren
Mercedes
RB (AlphaTauri)
Red Bull Racing
Williams
Red Bull Racing
Mercedes
Ferrari
Haas
Red Bull Racing
McLaren
Alpine
Ferrari
McLaren
Mercedes
Data Source
Performance Ranking
Team Championship Performan...
Tyre Compound Analysis
Country-wise Participation Anal...
Podium Achievements Final
Dashboard 1
Sheet 13
Dashboard 2
Dashboard 3
KPI
Story 1
Gyanvi Jain
27°C
Partly cloudy
Search
ENG
24%
6:58 PM
10/28/2026


F1 Race Intelligence | Formula
Ask Gemini
127.0.0.1:5000/#contact
School
F1 RACE INTELLIGENCE
Home
Project
Insights
Dashboard
Story
Contact
THE SCIENCE BEHIND THE SPEED
FORMULA 1
RACE
INTELLIGENCE
Every lap tells a story. Every second matters. Discover the data behind driver performance, team competition, race outcomes, and circuit strategy.
EXPLORE DASHBOARD ↗
DISCOVER THE PROJECT
DATA ANALYTICS · RACE STRATEGY · PERFORMANCE
01 / RACE ANALYTICS
27°C
Mostly cloudy
Search
ENG
39%
6:07 PM
10/28/2026